

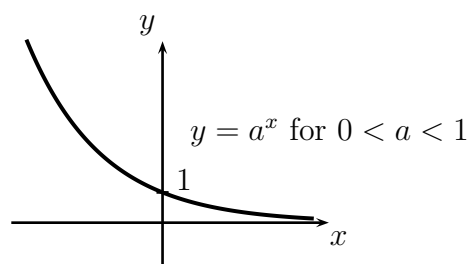
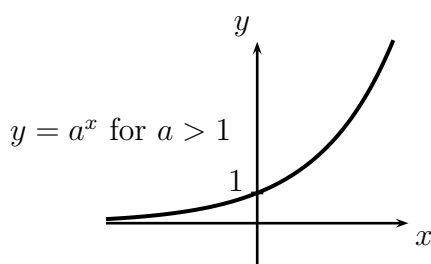
Notes for ‘The exponential function’

Important Ideas and Useful Facts:

- (i) Exponential functions: The simplest form of an exponential function f is

$$y = f(x) = a^x$$

where a is some fixed positive real number. When $a > 1$ the graph of f is sloping upwards. When $a < 1$ the graph of f is sloping downwards. When $a = 1$, the graph is the line $y = 1$, which is not an interesting case and usually excluded from discussion. In all cases the y -intercept is $y = 1$.



When $a \neq 1$, the domain of f is $\mathbb{R}$, the range is $(0, \infty)$, the graph satisfies the horizontal line test so f^{-1} exists (described later when discussing *logarithmic functions*), and the x -axis is a horizontal asymptote.

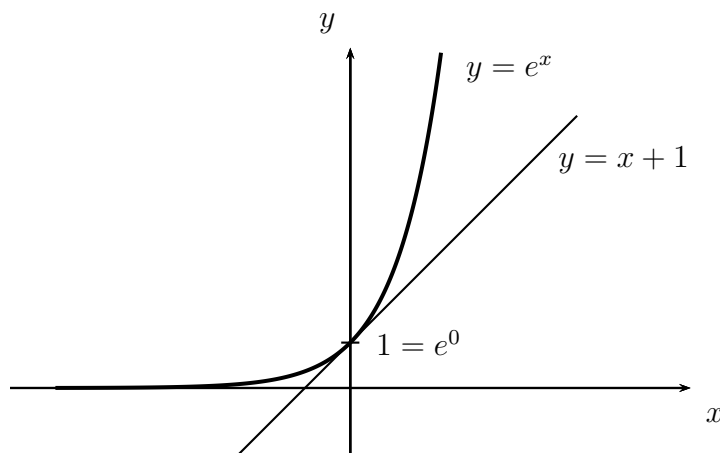
- (ii) Exponential laws: For all x and y , and for all positive a and b ,

$$a^x a^y = a^{x+y}, \quad \frac{a^x}{a^y} = a^{x-y}, \quad (a^x)^y = a^{xy}, \quad a^x b^x = (ab)^x.$$

- (iii) Euler's number: The real number

$$e = 2.718281828 \dots,$$

named after Leonhard Euler (1707-1783), is chosen to have the property that the tangent line to the curve $y = e^x$ at the y -intercept has slope 1.



This has important consequences when we discuss *derivatives* later. (In fact e^x will turn out to be its own derivative.) Euler's number e is irrational (though the proof is difficult).

Examples:

1. The following calculations involving simple numbers and integer exponents are consistent with the exponential laws given above:

$$2^2 \times 2^3 = (2 \times 2) \times (2 \times 2 \times 2) = 2^{2+3} = 2^5 = 32 ,$$

$$\frac{2^2}{2^3} = \frac{2 \times 2}{2 \times 2 \times 2} = \frac{\cancel{2} \times \cancel{2}}{\cancel{2} \times \cancel{2} \times 2} = 2^{2-3} = 2^{-1} = \frac{1}{2} ,$$

$$(2^2)^3 = (2 \times 2) \times (2 \times 2) \times (2 \times 2) = 2^{3 \times 2} = 2^{2 \times 3} = 2^6 = 64 ,$$

$$(2^2)(3^2) = (2 \times 2)(3 \times 3) = (2 \times 3) \times (2 \times 3) = 6 \times 6 = 6^2 = (2 \times 3)^2 = 36 .$$

2. The following calculations involving simple numbers and fractional powers employ the exponential laws, in several variations:

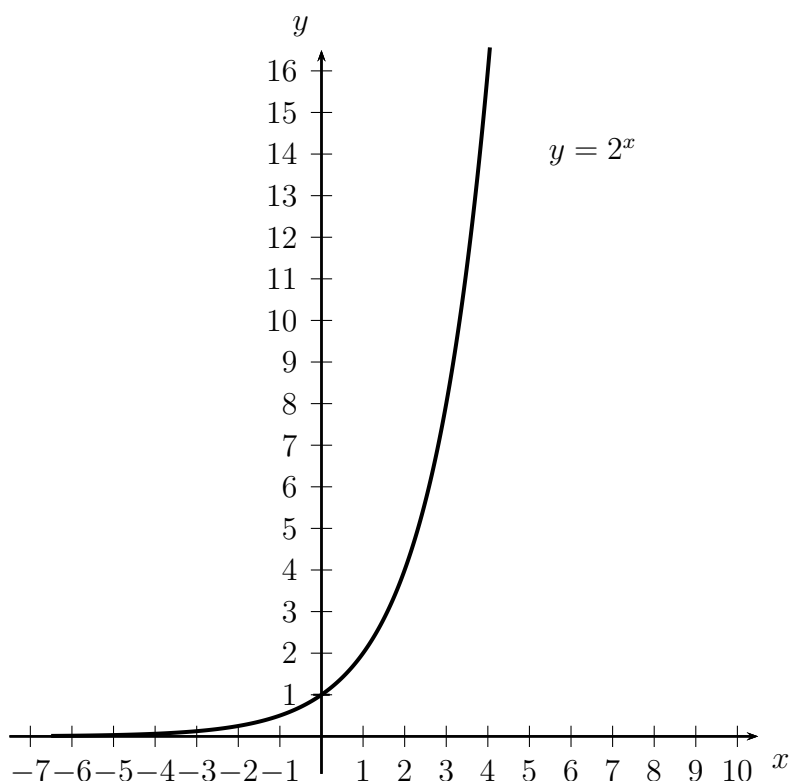
$$32^{4/5} = (32^{1/5})^4 = ((2^5)^{1/5})^4 = (2^{5/5})^4 = (2^1)^4 = 2^4 = 16 ,$$

$$32^{-3/5} = \frac{1}{32^{3/5}} = \frac{1}{(32^{1/5})^3} = \frac{1}{((2^5)^{1/5})^3} = \frac{1}{(2^{5/5})^3} = \frac{1}{2^3} = \frac{1}{8} ,$$

$$81^{3/4} = (9^2)^{3/4} = 9^{2 \times 3/4} = 9^{3/2} = (9^{1/2})^3 = 3^3 = 27 ,$$

$$\frac{2^{2.5} \times 3^{2.5}}{\sqrt{6}} = \frac{(2 \times 3)^{2.5}}{6^{1/2}} = \frac{6^{2.5}}{6^{0.5}} = 6^{2.5-0.5} = 6^2 = 36 .$$

3. Consider the graph of the exponential function $y = 2^x$, part of which is drawn below.



Notice that the x -axis in this diagram includes points up to and including $x = 10$, though the corresponding points on the curve shoot off into the distance.

In order to fit this diagram into the particular space chosen on the previous page, the scale is about 0.75 cm for each unit along the x and y -axes. To get an idea how steep this curve is, let's calculate how long the page would need to be so that (using the same scale) the diagram would include all the points on the curve up to and including $x = 10$. In other words, how long would the y -axis need to be to include the point on the curve with coordinates $(10, 2^{10})$?

With the present scale the y -axis would need to have length (measured from the origin)

$$2^{10} \times (0.75) \text{ cm} = 768 \text{ cm} ,$$

which is over seven and a half metres! This is an example of *exponential explosion*, explored in more detail later.